Shubham Kaushik

Data Systems Researcher | Software Engineer
shubhamkaushik.com | linkedin.com/in/shukham | github.com/shukham | kaushiks@brandeis.edu

Build storage engine internals in C++ and Rust: merge policies, memory layouts, and learning-based database tuning.
Four years of industry engineering and three years of research; published at VLDB, ICDE, TPCTC and DBTest.

EXPERIENCE

Ph.D. Researcher, **Smart & Scalable Data Systems Lab** @ Brandeis University, MA, USA **Jan 2024 - Present**
- Design and evaluate LSM-based storage engine internals such as merge policy, layouts, and write-buffer design.
- Apply learning-based methods to storage-engine configuration, from offline tuning to runtime adaptation.
- Develop benchmarking suites used to evaluate key-value stores under dynamically shifting workloads.

Software Engineer, **Scalability & Infrastructure Team** @ Kwalee, India **Mar 2022 - Aug 2022**
- Designed and shipped server-side leaderboard and loot-box systems integrated into three game releases.

Engineer - Information Security, **Information Security Automation** @ FIS Global, India **Jun 2021 - Mar 2022**
- Built an asynchronous IP-scanner to scan over 100K+ internet endpoints, replacing the team's manual spot-checks.
- Authored and merged integrations for DB2, GenericSQL and FireMon into Palo Alto's Cortex XSOAR [demisto/content].

Project Engineer, **Cloud Computing and Big Data Teams** @ Wipro Limited, India **Jul 2018 - Jun 2021**
- Built a microservice that pseudonymises PII in CSV, Excel and mainframe files, replacing a manual review of 1K+ files.
- Designed a real-time agent that flags anomalous hosts across 500+ enterprise systems using k-means clustering.

PROJECTS [SELECTED]

FluiDB — self-tuning storage engine [blog] **2025 - Present**
- Designing an LSM engine (prototype on RocksDB) that retunes itself as workloads shift, using reinforcement learning to search its configuration space, with an LLM acting as a DBA over prior tuning runs to narrow the search.

RangeReduce — query-driven compaction [PDF] [GitHub] **ICDE 2026**
- Designed a range query driven compaction strategy that reduce range query latency and the need for future compactions.
- Improves latency up to 18%, compaction debt up to 90%, data movement up to 12%, and space amplification up to 20%.

Tectonic → **TexBench** — key-value benchmarking suite [GitHub] **VLDB 2026, TPCTC 2025/26**
- Built *Tectonic*, a Rust key-value workload generator replacing YCSB: 2× higher throughput, 84% lower memory footprint.
- Developed *TexBench*, an LLM-powered layer on *Tectonic* that lets users specify benchmark workloads in plain English.
- Designed a Domain-Specific Language that restrict LLMs to generate eight edit commands to produce valid benchmarks.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Rust
Databases: RocksDB, PostgreSQL, Redis
Systems & Performance: Linux, perf, gdb, CMake, Docker
Machine Learning: Reinforcement Learning (RL), LLMs, DSL, Retrieval-Augmented Generation (RAG), PyTorch

PUBLICATIONS [SELECTED]

TexBench: Harnessing LLMs for Efficient Key-Value Benchmarking — **Kaushik** et al. TPCTC 2026
TexBench: A Unified Benchmarking Suite for Shifting Workloads (demo) — Chanda, **Kaushik** et al. VLDB 2026
RangeReduce: Query-Driven Compaction in LSM-Trees — **Kaushik** et al. ICDE 2026
Tectonic: Bridging Synthetic and Real-World Workloads — Alexander H. Ott, **Kaushik** et al. TPCTC 2025
Anatomy of the LSM Memory Buffer: Insights & Implications — **Kaushik** and Sarkar DBTest 2024

EDUCATION

Ph.D. in *Computer Science* (GPA: *3.98/4.0*) | **Brandeis University** **Jan 2024 - Present**
M.S. in *Computer Science*, w/ *Data-Centric Computing* (GPA: *3.88/4.0*) | **Boston University** **Sep 2022 - Dec 2023**
B.Tech. in *Computer Science & Engineering* | **Maharshi Dayanand University** **Jul 2014 - Jun 2018**

TALKS [SELECTED & INVITED]

RangeReduce: Query-Driven LSM Compactions @ MiDAS Seminar, Boston University, USA 04/2026
Range Queries and LSM-Trees: From Foes to Friends @ Salesforce, USA 10/2025

SERVICE

Program Committee, **VLDB 2027** · VLDB 2026 Shadow PC · Web Chair, NEDB Day 2025 · Reviewer, IEEE Big Data

OUTREACH

Volunteered at *VLDB 2026* · Judged and mentored at *HackMIT* · Mentored at *BostonHacks*